This document is intended to provide a basis for Midterm Senior Project Evaluations and is intended for use by the Faculty Coach, the Project Team, and any stakeholders, including the Project Sponsor. The evaluation criteria below are divided into two main areas: Process/Project, and Product. As it is early in the project, primary focus is placed on the Process/Project areas; later evaluations will focus much more heavily on product. Because each area is expansive and often includes entire classes covering them, a sample rubric, some questions, and general evaluation categories have been provided to increase understanding of what is expected throughout the project.

To use this document, go through each area and facilitate discussion. It is not necessary to complete each rubric or to fill in each question, but students should be encouraged to individually do so, meet as a group for comparison, and report back to the faculty coach and sponsor any results or conclusions they reach.

Process and Project

The tendency to focus on the product often leaves the process and the project out in the cold, especially in projects with set or compressed schedules. A well developed Software Engineering team knows that the returns paid within the Process and Project domains far outweigh the time and expenses that occur. Although the list below is not exhaustive, it includes some of the primary areas of importance.

Meetings and Communications Management

Sample Rubric: Meetings and Communications Management				
Meetings were held only at regular intervals or as a result of a crisis and did not include agendas or pre-planning	Meetings included standing agendas which were occasionally followed	Meetings included predetermined agendas, but the agenda was not used within the meeting or evaluated afterward. Included some items which did not necessitate a meeting	Meetings included predetermined agendas and were somewhat organized, but were occasionally allowed to stray from format. Content necessitated a meeting	Meetings were pre-organized, included prepared agendas, followed expected format, and did not include content that did not necessitate a meeting
Communication with stakeholders was reactive or crisis-driven, did not occur when expected, and did not clearly communicate expectations	Communication was conducted out of necessity but was rarely proactive and did not account for stakeholder expectation management	Communication was scheduled and typically took place. Expectation management was rarely contemplated	Communications with stakeholders were regular and generally proactive. Expectation management was occasionally used.	Communications with stakeholders were regular, proactive, and purpose-driven. Language was clear and included expectation management
In terms of customer education, team had a negative attitude towards the customer	Team blindly accepted customer inputs, did not convey own knowledge adequately	Team missed some occasions when customers needed education or	Team provided technical inputs to customer when needed or asked for	Team made sure at every stage to educate customer and provide/seek clarifications

and did not bother to assist them		clarifications were needed		
Comments: Sponsor agrees.				

1. Is the team conducting effective meetings? What can be changed to make the team meetings more productive?

The team is conducting effective meetings by meeting regularly at regular times. The agenda is followed and tasks are always pre-organized and follow an expected format.

2. How well has the team been communicating project progress to the sponsor? What regular communication does the team have with the sponsor? Has the team been maintaining this communication to the satisfaction of the sponsor? Were any adjustments needed in the communication over time? Were these changes initiated by the team or the sponsor?

Communication with the sponsor has been good, except for rare circumstances. There is a mechanism for the team to demo their work with the sponsor at the end of every sprint. In the past when this mechanism didn't work as the sponsor was not able to attend, it was difficult to keep the sponsor apprised of the project's progress.

3. Did the team need to provide technical input to the sponsor? How well did the team educate the customer in these areas? What mechanism did the team use?

The team did need to provide technical input to the sponsor, it seemed effective but that is ultimately up to the sponsor. The team used flow diagrams, wireframes, and use cases to communicate their input.

4. What mechanism does the team use to communicate with the faculty coach? Has communication with the coach been effective? Are there any trouble spots with the faculty coach communications? What can the team change for next term to make their communication to the faculty coach more effective? What can the faculty coach change to make his or her interaction with the team more effective?

The team uses email to communicate with the faculty coach, and communication has been effective. No trouble spots, and no changes were needed.

Sponsor/Stakeholder Communications

Comments:
Sponsor agrees with the team's assessment. Continue with our mechanisms.

Faculty Coach Communications

Comments:
1 meeting missed, but the sponsor was still able to provide feedback.

SE Department Personnel Communications and Coordination with Other Teams

Comments:

Time and Effort

Sample Rubric: Time and Effort				
Student effort was minimal, not meeting minimum requirements and putting the success of the project in crisis	Students effort only met minimum standards, and excuses or blame was often used to explain the effect of internal or external hampering factors	Student effort was adequate to meet project requirements, but was frequently hampered by both internal and external factors	Student effort was consistently better than average, even with negative internal factors. External factors occasionally affected effort levels	Students consistently put in full effort, even when internal or external factors have hampered progress
Comments: These meetings are great for learning about the goals of the project, the process, and working with stakeholders.				

1. Has the level of effort of individual team members been reflected in the output of the team as a whole?

The level of effort is reflected in the output of each member, it can be easily told from things like time tracking, agenda weekly points of interest, github pull requests, activity in their assigned team roles, and assigned tasks on the roadmap.

2. To what extent has thrashing (unproductive work) become a factor?

Thrashing has been extremely minimal, no produced artifacts were unnecessary and all work has been well focused on producing a comprehensively defined MVP.

Conformance to Schedule

Sample Rubric: Conformance to Schedule				
Did not demonstrate any schedule management and did not meet schedule requirements	Met minimal schedule requirements but showed no schedule management	Met most schedule requirements with little or no active schedule management	Met most or all schedule requirements with some active schedule management	Actively managed and met all schedule requirements
Comments: Sponsor indicated that the team needs to develop a mature process and understand that the team experiences unplanned events that can impact the velocity.				

1. Has the team met all project milestones to date? Which milestones, if any, were missed or were met ahead of schedule? What contributed to this schedule changes? What will the team do differently to ensure that future milestones are met?

To date, all project milestones have been met, and in most cases exceeded. The only milestone missed would be this assignment, and it has nothing to do with the team - the sponsor is on vacation and unable to meet at a time that fits with the team's schedule.

Change and Scope Management

Sample Rubric: Change and Scope Management				
Did not demonstrate awareness of issues as change or scope issues but rather as individual problems	Occasionally identified change or scope issues as such but often addressed them reactively as individual problems	Addressed change and scope when issues were presented with little or no root cause analysis	Addressed change and scope regularly but not always actively addressing change and scope issues before they occurred	Actively managed change and scope, seeking out and addressing issues before they affected the project. Used root cause analysis
Comments: Sponsor agrees.				

1. Has the scope of the project been considered at each milestone?

Scope has been a constant topic for the team leadership, and has been carefully managed to be as large as possible while still being achievable for the team.

Risk Management

Comments:
No comment.

Deliverables

Sample Rubric: Deliverables				
Deliverable was not present and not substituted with other efforts or deliverables	Deliverable was delivered with minimal effort and did not contribute significantly to the success of the project. Some obvious problems or inconsistencies were present, demonstrating little QA occurred	Deliverable met standards and was useful at introduction, but was not updated or useful further along in the project. Few obvious problems or inconsistencies were present	Deliverable was better than average and was used or enhanced irregularly, performing its desired function within the project. No obvious problems or inconsistencies were present	Deliverable performed as expected, was reviewed and enhanced at regular intervals, and would have consistently allowed someone not familiar with the project to become informed
Comments: No comment.				

1. How often do these artifacts get updated on the department project website?

Artifacts are currently not getting updated on the project website. The only place artifacts are visible is the T.R.A.I.L.S. repository.

2. Has the team had any issues with configuration management? How were these problems solved? What percentage of project artifacts is under configuration control?

No issues with configuration management outside of security leaks. The leaks were solved by responding as if it were an actual incident, preparing mitigation plans, and re-hiding the leaked information. As of now, 2-3 artifacts have been about configuration control.

Project Website

Comments:
No comment.

Time Tracking

Comments:
No comment.

Project Plan, Similar Project Deliverables

Comments:
No comment.

Risk Management Table

Comments:
No comment.

Quality

Sample Rubric: Quality				
Little attention was paid to initial requirements and the final product related only loosely to them	Requirements were created and used as a guideline for initial efforts, but were rarely addressed later in the project	Requirements management and quality control were performed irregularly, but there was an understanding of the link between quality and conformance to requirements	Conformance to functional requirements was a priority, but non-functional requirements often took a back seat, relating to quality control	Conformance to functional and non-functional requirements was clearly a priority and actively prevented scope creep and similar issues
A customer-view of quality was not considered throughout the project	Customer satisfaction was implied as a requirement but not effectively understood by the team. Treated as a requirement rather than a separate quality factor	The group understood that customer satisfaction was important, and occasionally addressed it, but focused primarily on requirements	Customer satisfaction was a priority, and some attention was placed to areas such as maintainability, but some imbalance occurred	Customer satisfaction and a long-term scope as a priority was demonstrated in each project step and properly balanced with conformance to requirements and scope

Comments:
Sponsor asks: Could we update the storycards and add acceptance criteria?

What should the sponsor have us do, what should he have the next team do?

1. How well has the team maintained quality control over the project artifacts? Have all artifacts been reviewed for adherence to quality standards? What is the review process used by the team?

Quality control over artifacts has been maintained mostly through reviewing with the coach after creation. There is not as much internal quality control as there could be. Most if not all artifacts have been reviewed with the coach to improve them, but there is not a formal internal review process being used, it is simply creation of the artifacts, and other team members review and suggest improvements as they also work on the document.

2. Was there a large requirement to learn the problem domain? What approach was used to gain domain expertise? What forms of support did you receive?

There was a very large requirement to know the problem domain, and learn it if it was unfamiliar. Due to the nature of the project already having multiple teams work on it, there was very little ability to choose specific technologies to use based on experience. The team ended up having to do most of the planning for the two main technologies, AWS / Frontend (React), by people with previous experience in something similar. Team members with more experience would help explain things to those with less so that everyone has knowledge on the problem domain. Most of the experience gained was through the form of dedicated research, and finding and sharing guides that discuss the things that we need to do.

3. Has the work accomplished to date satisfied the project sponsor? Team's opinion? Sponsor's opinion? Were there any weak spots in this regard?

The work accomplished to date has likely satisfied the sponsor, but there really hasn't been too much visible progress since there was so much initial planning and refactoring, things that the sponsor wouldn't see many visible changes for on the application itself.

Process Creation and Improvement

Sample Rubric: Process Creation and Improvement				
Process is only used when clear best-practice standards exist or an external process must be followed	Processes have been created and are usually followed, but are rarely improved.	Processes have been created and are consistently followed, but are rarely improved. Crisis situations occasionally trigger process improvement	Processes are always followed and occasionally improved. Root cause analysis may be a byproduct but is not actively pursued	Active process improvement is addressed regularly and processes are evaluated/root cause analysis is used regularly
Metrics were not planned or used	Metrics were used only as required by	Metrics were regularly collected, but	Metrics were regularly collected	Metrics were properly planned,

throughout the project	faculty coach or project guidelines, but were rarely used and only tracked by necessity	inconsistently used, especially relating to process	and evaluated/used as intended in process and product domains.	implemented, proved useful to stakeholders, and were used for process improvement
Comments: The sponsor agrees.				

1. What is your process methodology? Has this been clearly outlined to your sponsor and received the sponsor's approval? How is the process documented?

Our process methodology is Agile (Scrum). The process was clearly defined to the sponsor and was recommended by the sponsor. Process methodology was approved and well documented.

2. What mechanisms is the team using to track project progress? How well has the team tracked its project progress?

The team is using Github Projects to track the progress of the project. Within the project are epics, stories, and features that the team must implement to complete an MVP. If MVP features are completed with time to spare, then there are also stretch goals for the team to hit. The team has tracked their progress effectively by always updating information as it happens, linking pull requests to issues and keeping others updated in meetings.

3. What is the set of metrics that the team is tracking? Has the team gathered these metrics on a consistent basis? What has the team learned from the review of these metrics?

The team is tracking requirements volatility, task slippage, and effort estimation accuracy. These metrics have been gathered on a semi-consistent basis. The team has learned that this project requires many extra steps to complete seemingly simple tasks for other teams, but that the team also completes tasks much faster than other teams - and everything evens out. We also found that a few team members spend much more time than others, but also everyone spends much more time than is required.

Team Effectiveness

Sample Rubric: Team Effectiveness				
Team fractured, unable to work together constructively.	Team had some significant frictions and lack of mutual cooperation.	Team had some significant disconnects and frictions, but worked through them.	Team worked reasonably well together, no problems visible.	Team has achieved synergy, allocated work well, and worked together as a cohesive whole.
Comments: The sponsor agrees with this.				

1. Was the team required to adopt new technologies? What were these technologies? What approach did the team use for selecting the appropriate technology for the project? Did the sponsor provide any support for learning these technologies? How well did the team ramp up on the new technologies and begin to apply them effectively?

The team was not required to adopt new technologies, however for some Amazon AWS was a new technology. We offset some inexperienced team members by utilizing other team members' skills in those areas. Additionally, there was a mobile component to the project that was decided to be beyond the scope of an MVP.

2. Is this an effective team? What has been contributing to and detracting from the team's effectiveness? What are the team's weak points? What are the team's strong points? What changes can the team make for next term that will make it more effective?

This is a highly effective team. The team members each have a strong grasp of their responsibilities and are comfortable with being held accountable for their artifacts. The team is weak at mobile development and engineering security. The team is strong at task completion, AWS, React, architecture/design, and documentation.

Product

Although Process and Project are important, the product is ultimately why projects are initiated and followed through to completion. An effective Software Engineer maintains a steady focus on the product, and understands the scope of what it entails. Although the list below is not exhaustive, and remains short for the first evaluation, it includes some of the primary areas of importance.

Requirements

Sample Rubric: Requirements				
Requirements have not been collected beyond initial project charter. Several key requirements missed	Requirements were collected somewhat effectively, but some key requirements were overlooked initially	Requirements were collected exhaustively, but some were not clarified adequately	Requirements were actively elicited, documented, and understood. Some traceability issues may be present	All functional requirements have been proactively collected and managed. Documentation and traceability is clear and defined
Comments: How do you get the map feature done? That is the important feature the sponsor wants completed. The sponsor is worried about the map view not getting done.				

1. Did the team prepare all the documentation artifacts requested by your faculty coach and sponsor? Were these documents carefully inspected prior to delivery? How would you assess the quality of the document artifacts?

   The team was able to provide all documentation requested from the coach and the sponsor. The documents were carefully inspected by at least half of the team members, as well as the coach. Every artifact submitted was high quality and each one that was prepared has a specific purpose and goal.

2. How well did the team elicit the requirements? Are the requirements fully specified at this point? What approaches were used to elicit the requirements? Were key requirements missed? What methodology was used to document and validate the project requirements?

The team did a very thorough analysis of the original requirement document, which eventually led to a revised requirements document that had additions, edits, and deletions from the original requirements document. Currently, the requirements are fully specified and reviewed with the sponsor. The team's initial meeting with the sponsor was to review the requirements document, discuss the project's history, and determine if any requirements were able to be reconsidered. The scope of the project is very large and the team tried to minimize the scope in order to meet the MVP. No key requirements were missed, a full conversion to mobile was added as a stretch goal for later usage. An agile methodology was used to gather requirements and plan out manageable amounts of work given those requirements. The methodology used to document and validate the project's requirements is to create architectural and behavioural models that are derived from requirements and will be created with a level of precision and unambiguity that enables a clean mapping from architectural entities to implementation entities in the chosen solution domain (AWS/React).

3. Products need to be designed within guidelines and constraints appropriate for each project. It is also important to consider the impacts of the products that are designed. In the following categories discuss the constraints and impacts that have a bearing on your project. Note that there may be one or two categories that have no bearing on your project but your project is probably affected by almost all of these.
   a. Economic issues
      Resource availability in order to acquire more devices due to inflation. The sponsor must be able to afford the maintenance of the application.
   b. Environmental issues
      Sending devices into the wilderness. If a device was lost or broken, it would lead to electronic waste
   c. Social issues

The project needs to be aware that its affecting wildlife habitats and needs to be hardened for that purpose.

d. Political issues
   N/A
e. Ethical issues
   People might not want their hike tracked by the system (even if it's just a counter)
f. Health and safety
   Trail counters might be placed in dangerous spots, or may require maintenance in harsh environments
g. Manufacturability
   Recreating trail counter devices and having them on every trail in the system
h. Sustainability
   Very minor impact on the trail managers who might need to continue checking devices after they are sent to trails

4. What industry and engineering standards must your project adhere to? Were these new standards that the team had to learn? Did your sponsor provide you support for understanding these standards? Did you have to educate your sponsor about these standards?

The project must adhere to the quality standards set for RIT students, and should be thinking about regulations for software that is run and maintained by NY state. These are standards that the team and the sponsor must learn.

Future Considerations:

Future evaluations will include heavier evaluation in the areas of Design, Implementation and Testing. What are the implications of these at the present point in the project?

The implications of this is that we need to be sure that our design, implementation, and testing is well defined, communicated, documented, and executed.

What can the team do to make the transition from this team to the next more smoothly?